ThetaSwap: Adverse Competition Oracle

The Risk Nobody Hedges

Sophisticated LPs concentrate fee revenue away from passive participants. This is **adverse competition** — a distinct, unhedged risk.

- **Impermanent loss (IL)** — tracks price divergence between pool assets. Options and structured products hedge IL.
- **Loss-versus-rebalancing (LVR)** — tracks adverse selection cost from informed traders. MEV-aware products address LVR.
- **Fee concentration risk** — tracks the *competition structure* among LPs for fee revenue. **Nothing hedges this today.**

A Passive LP's Experience

1. **Alice deposits liquidity** — expects $1/N$ fee share (Ma & Crapis equilibrium).
2. **JIT provider enters** — mints a tightly concentrated position for a single block, captures $x_k \gg 1/N$.
3. **Alice's fee rate drops** — effective fee rate $\hat{\phi}_i = x_i \cdot \phi_{\text{pool}}$ falls below her exit threshold (Capponi & Zhu, 2024).
4. **Alice exits earlier** — adverse competition pushes her out before she would have left voluntarily.

In 41 days of ETH/USDC data

Real A_T exceeds the null by $2.65\times$. 63% of days show positive deviation ($\Delta^+ > 0$).

Quadratic Hazard Model

$$\boxed{P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \Delta_{t-\ell} + \beta_2 \Delta_{t-\ell}^2 + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)}$$

- Negative β_1 + positive β_2 = **inverted-U**
- Below turning point: concentration is *protective* (shelter regime)
- Above turning point: concentration *drives exits* (Capponi regime)

Sample: $n = 3{,}365$ position-days · 597 exits · 40 day-clusters · ETH/USDC 30bps

Empirical Results

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	$\mathbf{-23.18}$	**0.012**	$\mathbf{+129.20}$	**0.030**	**0.090**
2	-43.42	*0.016*	$+226.92$	*0.065*	*0.096*
3	$\mathbf{-32.44}$	**0.001**	$\mathbf{+205.34}$	**0.004**	**0.079**

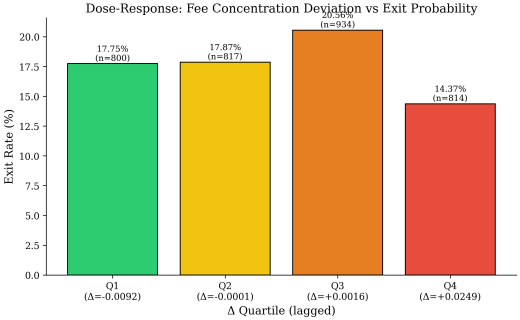
- **Lags 1 and 3**: both coefficients significant at 5%
- **Lag 2**: linear significant, quadratic marginal ($p = 0.065$)
- Signal **dissipates at lags 5–7** — LPs react to recent shocks

The Inverted-U

Two regimes:

1. **Shelter** ($\Delta < 0.09$)
   Mild concentration is protective — elevated volume benefits all LPs.

2. **Capponi** ($\Delta > 0.09$)
   Extreme concentration drives exits — passive LP fee rate drops below exit threshold.

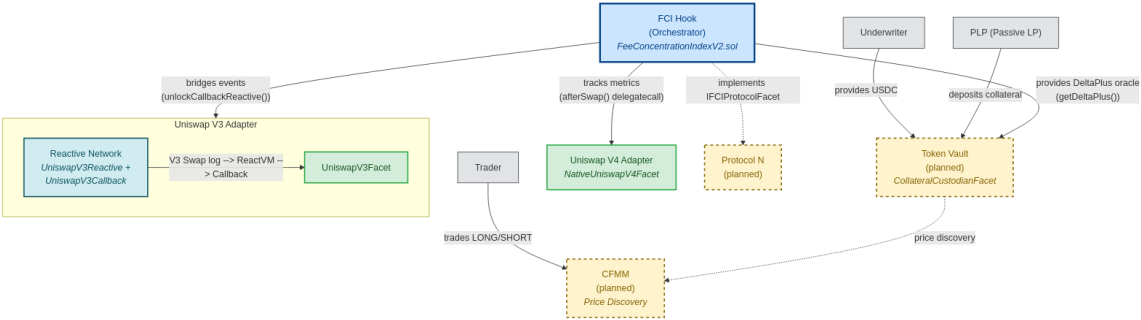


Turning point:

$$\Delta^* = \frac{-\beta_1}{2\beta_2} = \frac{23.18}{2 \times 129.20} = 0.090$$

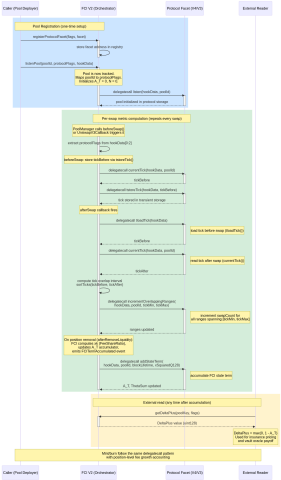
At $\Delta = 0.15$: marginal effect $= 2.27$

Architecture



FCI Hook at center — protocol adapters (V3, V4) feed swap/mint/burn events, Reactive Network bridges metrics cross-chain, Vault + CFMM settle downstream.

Cross-Protocol Modularity



`listenPool()` → swap/mint/burn events → metric update → Δ^+ derivation.

Live Demo

Run the integration test

```
forge test --match-path
"test/fee-concentration-index-v2/protocols/uniswapV4/
NativeV4FeeConcentrationIndex.integration.t.sol" -vv
```

What to observe:

1. Swap updates A_T — fee concentration index tracks each trade
2. Mint increments N — new positions enter the pool
3. Burn triggers Δ^+ derivation — exit computes concentration deviation
4. All metrics computed on-chain in a single V4 hook callback

What's Next

Built

- FCI oracle live on Uniswap V4
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value

Next

- Linearized power-squared CFMM (trading function complete in spec)
- Vault settlement mechanism

The mathematical specification is complete, the oracle tracks metrics across protocols, and the next milestone delivers the full insurance product.